ORF309 Homework 6 - Question 1a

Question 1a: Expected Lifetime

Problem Setup

We are modeling a person's lifetime L where the person may experience a heart attack at random time T. The heart attack increases their risk of dying.

Variable Definitions

- L = Lifetime of the person (random variable we want to find $E[L]$)
- T = Time when heart attack occurs (random variable), where $T \sim \text{Expon}(\mu)$
- $h(t)$ = Hazard rate at time t (instantaneous risk of dying)
  - $h(t) = a$ for $t < T$ (before heart attack)
  - $h(t) = b$ for $t \geq T$ (after heart attack, typically $b > a$)
- τ = A specific value that T could take (used when conditioning)
- a, b, μ = Given constants (parameters)

Strategy

Since L depends on the random variable T, we use the **Law of Total Expectation**:

$$E[L] = \int_0^\infty E[L \mid T = \tau] \cdot f_T(\tau)\, d\tau$$

where $f_T(\tau) = \mu e^{-\mu\tau}$ is the probability density of T.

Intuition: First compute the expected lifetime for each possible heart attack time τ, then average over all possible τ values weighted by their probability.

Step 1: Find $E[L \mid T = \tau]$

Given that the heart attack occurs at time $T = \tau$, we need to find the expected lifetime.

Step 1a: Cumulative Hazard $H(t)$

The cumulative hazard is the total accumulated risk from birth to time t:

$$H(t) = \int_0^t h(s)\,ds$$

For $t < \tau$ (before heart attack):

$$H(t) = \int_0^t a\,ds = at$$

For $t \geq \tau$ (after heart attack):

$$H(t) = \int_0^\tau a\,ds + \int_\tau^t b\,ds = a\tau + b(t-\tau)$$

Step 1b: Survival Function $S(t)$

The survival function is the probability of surviving past time t:

$$S(t) = e^{-H(t)}$$

For $t < \tau$:

$$S(t) = e^{-at}$$

For $t \geq \tau$:

$$S(t) = e^{-a\tau - b(t-\tau)}$$

Step 1c: Compute $E[L \mid T = \tau]$

Using the formula from survival analysis:

$$E[L \mid T = \tau] = \int_0^\infty S(t)\,dt$$

Split the integral at τ:

$$E[L \mid T = \tau] = \int_0^\tau e^{-at}\,dt + \int_\tau^\infty e^{-a\tau - b(t-\tau)}\,dt$$

First integral:

$$\int_0^\tau e^{-at}\,dt = \left[-\frac{1}{a}e^{-at}\right]_0^\tau = \frac{1}{a}(1 - e^{-a\tau})$$

Second integral (substitute $v = t - \tau$, so $dv = dt$):

$$\begin{aligned}
\int_\tau^\infty e^{-a\tau - b(t-\tau)}\,dt &= e^{-a\tau}\int_0^\infty e^{-bv}\,dv \\
&= e^{-a\tau}\left[-\frac{1}{b}e^{-bv}\right]_0^\infty \\
&= e^{-a\tau}\cdot\frac{1}{b} = \frac{e^{-a\tau}}{b}
\end{aligned}$$

Combining both integrals:

$$
\begin{aligned}
E[L \mid T = \tau] &= \frac{1}{a}(1 - e^{-a\tau}) + \frac{e^{-a\tau}}{b} \\
&= \frac{1}{a} - \frac{e^{-a\tau}}{a} + \frac{e^{-a\tau}}{b} \\
&= \frac{1}{a} + e^{-a\tau}\left(\frac{1}{b} - \frac{1}{a}\right)
\end{aligned}
$$

Step 2: Average Over All Possible τ Values

Now we apply the Law of Total Expectation. Since $T \sim \text{Expon}(\mu)$ with density $f_T(\tau) = \mu e^{-\mu\tau}$:

$$
\begin{aligned}
E[L] &= \int_0^\infty E[L \mid T = \tau] \cdot \mu e^{-\mu\tau}\, d\tau \\
&= \int_0^\infty \left[\frac{1}{a} + e^{-a\tau}\left(\frac{1}{b} - \frac{1}{a}\right)\right] \mu e^{-\mu\tau}\, d\tau
\end{aligned}
$$

Split into two integrals:

$$
\begin{aligned}
E[L] &= \frac{\mu}{a}\int_0^\infty e^{-\mu\tau}\, d\tau + \mu\left(\frac{1}{b} - \frac{1}{a}\right)\int_0^\infty e^{-a\tau}e^{-\mu\tau}\, d\tau \\
&= \frac{\mu}{a}\int_0^\infty e^{-\mu\tau}\, d\tau + \mu\left(\frac{1}{b} - \frac{1}{a}\right)\int_0^\infty e^{-(a+\mu)\tau}\, d\tau
\end{aligned}
$$

First integral:

$$
\int_0^\infty e^{-\mu\tau}\, d\tau = \left[-\frac{1}{\mu}e^{-\mu\tau}\right]_0^\infty = \frac{1}{\mu}
$$

Second integral:

$$
\int_0^\infty e^{-(a+\mu)\tau}\, d\tau = \left[-\frac{1}{a+\mu}e^{-(a+\mu)\tau}\right]_0^\infty = \frac{1}{a+\mu}
$$

Final Simplification

Substituting back:

$$
\begin{aligned}
E[L] &= \frac{\mu}{a} \cdot \frac{1}{\mu} + \mu\left(\frac{1}{b} - \frac{1}{a}\right) \cdot \frac{1}{a+\mu} \\
&= \frac{1}{a} + \frac{\mu}{b(a+\mu)} - \frac{\mu}{a(a+\mu)}
\end{aligned}
$$

To combine terms, find common denominator $a(a+\mu)$:

$$\begin{aligned}
E[L] &= \frac{a+\mu}{a(a+\mu)} + \frac{\mu}{b(a+\mu)} - \frac{\mu}{a(a+\mu)} \\
&= \frac{(a+\mu)-\mu}{a(a+\mu)} + \frac{\mu}{b(a+\mu)} \\
&= \frac{a}{a(a+\mu)} + \frac{\mu}{b(a+\mu)} \\
&= \frac{1}{a+\mu} + \frac{\mu}{b(a+\mu)} \\
&= \frac{b+\mu}{b(a+\mu)}
\end{aligned}$$

Answer

$$\boxed{E[L] = \frac{b+\mu}{b(a+\mu)}}$$